

Chapter - 11 Constructions

Exercise 11.1

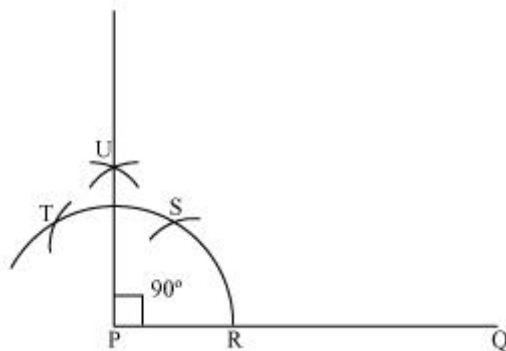
Question 1:

Construct an angle of 90° at the initial point of a given ray and justify the construction.

Answer:

The below given steps will be followed to construct an angle of 90° .

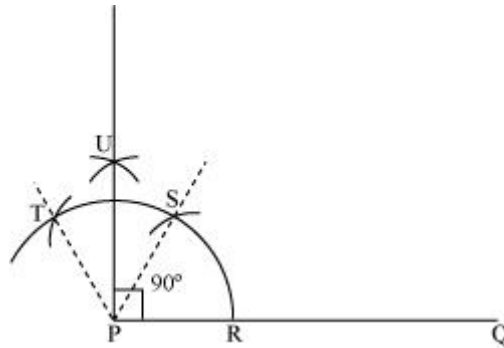
- (i) Take the given ray PQ. Draw an arc of some radius taking point P as its centre, which intersects PQ at R.
- (ii) Taking R as centre and with the same radius as before, draw an arc intersecting the previously drawn arc at S.
- (iii) Taking S as centre and with the same radius as before, draw an arc intersecting the arc at T (see figure).
- (iv) Taking S and T as centre, draw an arc of same radius to intersect each other at U.
- (v) Join PU, which is the required ray making 90° with the given ray PQ.



Justification of Construction:

We can justify the construction, if we can prove $\angle UPQ = 90^\circ$.

For this, join PS and PT.



We have, $\angle SPQ = \angle TPS = 60^\circ$. In (iii) and (iv) steps of this construction, PU was drawn as the bisector of $\angle TPS$.

$$\therefore \angle UPS = \frac{1}{2} \angle TPS = \frac{1}{2} \times 60^\circ = 30^\circ$$

Also, $\angle UPQ = \angle SPQ + \angle UPS$

$$= 60^\circ + 30^\circ$$

$$= 90^\circ$$

Question 2:

Construct an angle of 45° at the initial point of a given ray and justify the construction.

Answer:

The below given steps will be followed to construct an angle of 45° .

(i) Take the given ray PQ. Draw an arc of some radius taking point P as its centre, which intersects PQ at R.

(ii) Taking R as centre and with the same radius as before, draw an arc intersecting the previously drawn arc at S.

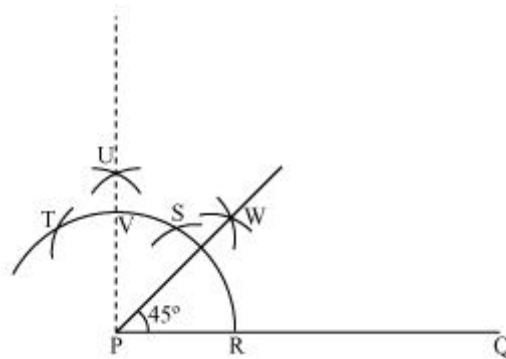
(iii) Taking S as centre and with the same radius as before, draw an arc intersecting the arc at T (see figure).

(iv) Taking S and T as centre, draw an arc of same radius to intersect each other at U.

(v) Join PU. Let it intersect the arc at point V.

(vi) From R and V, draw arcs with radius more than $\frac{1}{2}RV$ to intersect each other at W. Join PW.

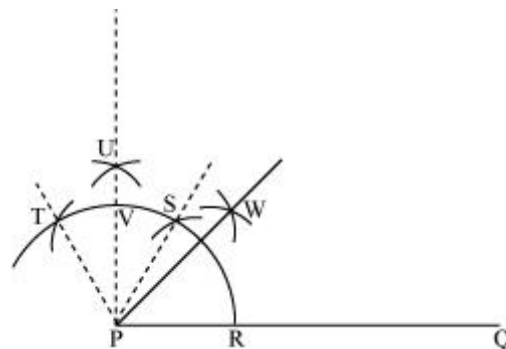
PW is the required ray making 45° with PQ.



Justification of Construction:

We can justify the construction, if we can prove $\angle WPQ = 45^\circ$.

For this, join PS and PT.



We have, $\angle SPQ = \angle TPS = 60^\circ$. In (iii) and (iv) steps of this construction, PU was drawn as the bisector of $\angle TPS$.

$$\therefore \angle UPS = \frac{1}{2} \angle TPS = \frac{60^\circ}{2} = 30^\circ$$

$$\text{Also, } \angle UPQ = \angle SPQ + \angle UPS$$

$$= 60^\circ + 30^\circ$$

$$= 90^\circ$$

In step (vi) of this construction, PW was constructed as the bisector of $\angle UPQ$.

$$\therefore \angle WPQ = \frac{1}{2} \angle UPQ = \frac{90^\circ}{2} = 45^\circ$$

Question 3:

Construct the angles of the following measurements:

$$(i) 30^\circ \quad (ii) 22\frac{1}{2}^\circ \quad (iii) 15^\circ$$

Answer:

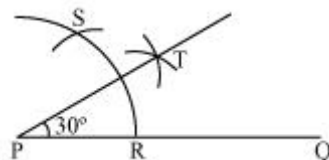
$$(i) 30^\circ$$

The below given steps will be followed to construct an angle of 30° .

Step I: Draw the given ray PQ. Taking P as centre and with some radius, draw an arc of a circle which intersects PQ at R.

Step II: Taking R as centre and with the same radius as before, draw an arc intersecting the previously drawn arc at point S.

Step III: Taking R and S as centre and with radius more than $\frac{1}{2} RS$, draw arcs to intersect each other at T. Join PT which is the required ray making 30° with the given ray PQ.



$$(ii) 22\frac{1}{2}^\circ$$

The below given steps will be followed to construct an angle of $22\frac{1}{2}^\circ$.

(1) Take the given ray PQ. Draw an arc of some radius, taking point P as its centre, which intersects PQ at R.

(2) Taking R as centre and with the same radius as before, draw an arc intersecting the previously drawn arc at S.

(3) Taking S as centre and with the same radius as before, draw an arc intersecting the arc at T (see figure).

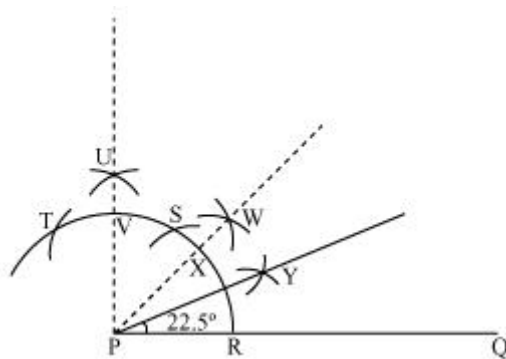
(4) Taking S and T as centre, draw an arc of same radius to intersect each other at U.

(5) Join PU. Let it intersect the arc at point V.

(6) From R and V, draw arcs with radius more than $\frac{1}{2}RV$ to intersect each other at W. Join PW.

(7) Let it intersect the arc at X. Taking X and R as centre and radius more than $\frac{1}{2}RX$, draw arcs to intersect each other at Y.

Join PY which is the required ray making $22\frac{1}{2}^\circ$ with the given ray PQ.



(iii) 15°

The below given steps will be followed to construct an angle of 15° .

Step I: Draw the given ray PQ. Taking P as centre and with some radius, draw an arc of a circle which intersects PQ at R.

Step II: Taking R as centre and with the same radius as before, draw an arc intersecting the previously drawn arc at point S.

Step III: Taking R and S as centre and with radius more than $\frac{1}{2}$ RS, draw arcs to intersect each other at T. Join PT.

Step IV: Let it intersect the arc at U. Taking U and R as centre and with radius more than $\frac{1}{2}$ RU, draw an arc to intersect each other at V. Join PV which is the required ray making 15° with the given ray PQ.

Question 4:

Construct the following angles and verify by measuring them by a protractor:

(i) 75° (ii) 105° (iii) 135°

Answer:

(i) 75°

The below given steps will be followed to construct an angle of 75° .

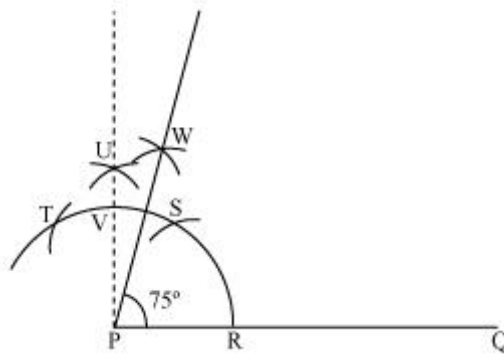
(1) Take the given ray PQ. Draw an arc of some radius taking point P as its centre, which intersects PQ at R.

(2) Taking R as centre and with the same radius as before, draw an arc intersecting the previously drawn arc at S.

(3) Taking S as centre and with the same radius as before, draw an arc intersecting the arc at T (see figure).

(4) Taking S and T as centre, draw an arc of same radius to intersect each other at U.

(5) Join PU. Let it intersect the arc at V. Taking S and V as centre, draw arcs with radius more than $\frac{1}{2}$ SV. Let those intersect each other at W. Join PW which is the required ray making 75° with the given ray PQ.



The angle so formed can be measured with the help of a protractor. It comes to be 75° .

(ii) 105°

The below given steps will be followed to construct an angle of 105° .

(1) Take the given ray PQ. Draw an arc of some radius taking point P as its centre, which intersects PQ at R.

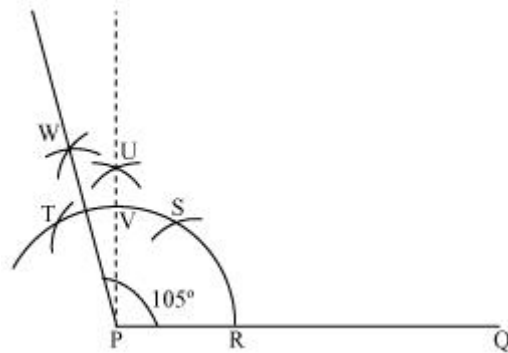
(2) Taking R as centre and with the same radius as before, draw an arc intersecting the previously drawn arc at S.

(3) Taking S as centre and with the same radius as before, draw an arc intersecting the arc at T (see figure).

(4) Taking S and T as centre, draw an arc of same radius to intersect each other at U.

(5) Join PU. Let it intersect the arc at V. Taking T and V as centre, draw arcs with

radius more than $\frac{1}{2}$ TV. Let these arcs intersect each other at W. Join PW which is the required ray making 105° with the given ray PQ.



The angle so formed can be measured with the help of a protractor. It comes to be 105° .

(iii) 135°

The below given steps will be followed to construct an angle of 135° .

(1) Take the given ray PQ. Extend PQ on the opposite side of Q. Draw a semi-circle of some radius taking point P as its centre, which intersects PQ at R and W.

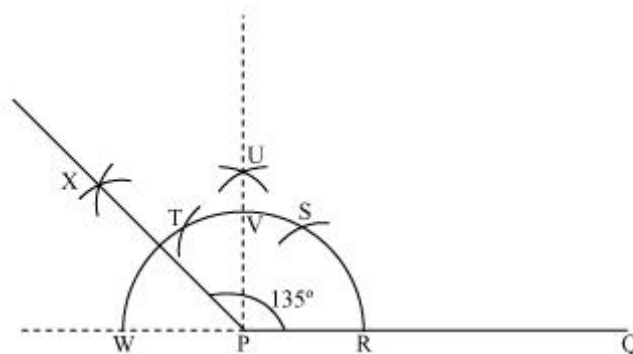
(2) Taking R as centre and with the same radius as before, draw an arc intersecting the previously drawn arc at S.

(3) Taking S as centre and with the same radius as before, draw an arc intersecting the arc at T (see figure).

(4) Taking S and T as centre, draw an arc of same radius to intersect each other at U.

(5) Join PU. Let it intersect the arc at V. Taking V and W as centre and with radius

more than $\frac{1}{2} VW$, draw arcs to intersect each other at X. Join PX, which is the required ray making 135° with the given line PQ.



The angle so formed can be measured with the help of a protractor. It comes to be 135° .

Question 5:

Construct an equilateral triangle, given its side and justify the construction

Answer:

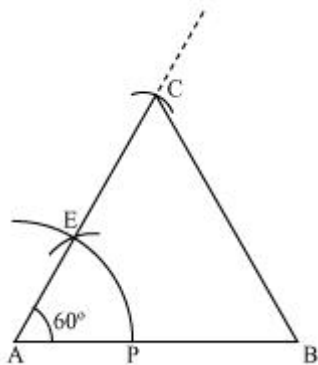
Let us draw an equilateral triangle of side 5 cm. We know that all sides of an equilateral triangle are equal. Therefore, all sides of the equilateral triangle will be 5 cm. We also know that each angle of an equilateral triangle is 60° .

The below given steps will be followed to draw an equilateral triangle of 5 cm side.

Step I: Draw a line segment AB of 5 cm length. Draw an arc of some radius, while taking A as its centre. Let it intersect AB at P.

Step II: Taking P as centre, draw an arc to intersect the previous arc at E. Join AE.

Step III: Taking A as centre, draw an arc of 5 cm radius, which intersects extended line segment AE at C. Join AC and BC. $\triangle ABC$ is the required equilateral triangle of side 5 cm.



Justification of Construction:

We can justify the construction by showing ABC as an equilateral triangle i.e., $AB = BC = AC = 5$ cm and $\angle A = \angle B = \angle C = 60^\circ$.

In $\triangle ABC$, we have $AC = AB = 5$ cm and $\angle A = 60^\circ$.

Since $AC = AB$,

$\angle B = \angle C$ (Angles opposite to equal sides of a triangle)

In $\triangle ABC$,

$\angle A + \angle B + \angle C = 180^\circ$ (Angle sum property of a triangle)

$$\Rightarrow 60^\circ + \angle C + \angle C = 180^\circ$$

$$\Rightarrow 60^\circ + 2 \angle C = 180^\circ$$

$$\Rightarrow 2 \angle C = 180^\circ - 60^\circ = 120^\circ$$

$$\Rightarrow \angle C = 60^\circ$$

$$\therefore \angle B = \angle C = 60^\circ$$

We have, $\angle A = \angle B = \angle C = 60^\circ \dots (1)$

$$\Rightarrow \angle A = \angle B \text{ and } \angle A = \angle C$$

$\Rightarrow BC = AC \text{ and } BC = AB$ (Sides opposite to equal angles of a triangle)

$$\Rightarrow AB = BC = AC = 5 \text{ cm } \dots (2)$$

From equations (1) and (2), $\triangle ABC$ is an equilateral triangle.

Exercise 11.2

Question 1:

Construct a triangle ABC in which $BC = 7 \text{ cm}$, $\angle B = 75^\circ$ and $AB + AC = 13 \text{ cm}$.

Answer:

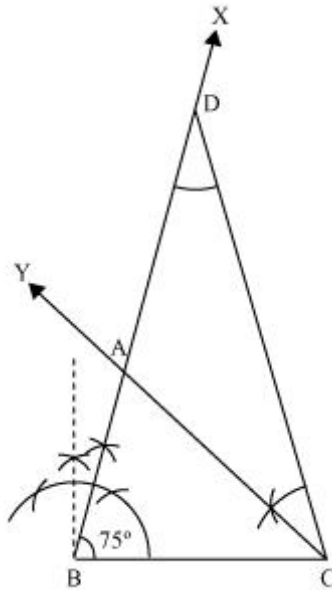
The below given steps will be followed to construct the required triangle.

Step I: Draw a line segment BC of 7 cm. At point B, draw an angle of 75° , say $\angle XBC$.

Step II: Cut a line segment BD = 13 cm (that is equal to $AB + AC$) from the ray BX.

Step III: Join DC and make an angle DCY equal to $\angle BDC$.

Step IV: Let CY intersect BX at A. $\triangle ABC$ is the required triangle.



Question 2:

Construct a triangle ABC in which $BC = 8$ cm, $\angle B = 45^\circ$ and $AB - AC = 3.5$ cm.

Answer:

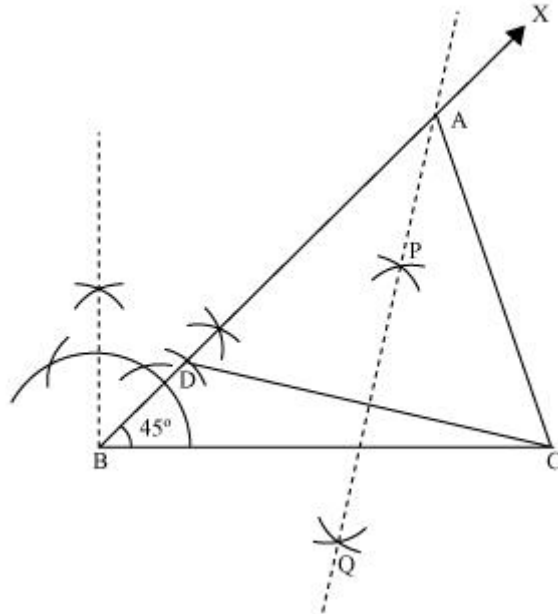
The below given steps will be followed to draw the required triangle.

Step I: Draw the line segment $BC = 8$ cm and at point B, make an angle of 45° , say $\angle XBC$.

Step II: Cut the line segment $BD = 3.5$ cm (equal to $AB - AC$) on ray BX.

Step III: Join DC and draw the perpendicular bisector PQ of DC.

Step IV: Let it intersect BX at point A. Join AC. $\triangle ABC$ is the required triangle.



Question 3:

Construct a triangle PQR in which $QR = 6$ cm, $\angle Q = 60^\circ$ and $PR - PQ = 2$ cm

Answer:

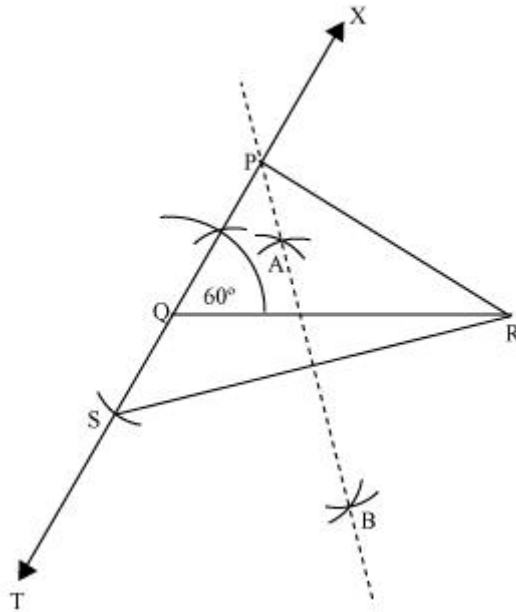
The below given steps will be followed to construct the required triangle.

Step I: Draw line segment QR of 6 cm. At point Q, draw an angle of 60° , say $\angle XQR$.

Step II: Cut a line segment QS of 2 cm from the line segment QT extended in the opposite side of line segment XQ. (As $PR > PQ$ and $PR - PQ = 2$ cm). Join SR.

Step III: Draw perpendicular bisector AB of line segment SR. Let it intersect QX at point P. Join PQ, PR.

$\triangle PQR$ is the required triangle.



Question 4:

Construct a triangle XYZ in which $\angle Y = 30^\circ$, $\angle Z = 90^\circ$ and $XY + YZ + ZX = 11$ cm.

Answer:

The below given steps will be followed to construct the required triangle.

Step I: Draw a line segment AB of 11 cm.

(As $XY + YZ + ZX = 11$ cm)

Step II: Construct an angle, $\angle PAB$, of 30° at point A and an angle, $\angle QBA$, of 90° at point B.

Step III: Bisect $\angle PAB$ and $\angle QBA$. Let these bisectors intersect each other at point X.

Step IV: Draw perpendicular bisector ST of AX and UV of BX.

Step V: Let ST intersect AB at Y and UV intersect AB at Z.

Join XY, XZ .

$\triangle XYZ$ is the required triangle.

$\triangle ABC$ is the required triangle.

